

Transdisciplinary Innovation: Connecting Ideas from Professional and Community Networks

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Abstract. Transdisciplinary innovation - what is it and how does it work? In this paper, the way disparate professional and community actors may work together is considered, drawing on case study data from three different Australian-Indian academic research collaborations. One considered food sector SME innovation practice in the two countries and the other two considered the deployment in India of radical technologies developed by international teams to deliver social benefits. The collection of knowledge artifacts from disparate sources was the norm. Implementation of an innovative idea or technology application commonly involved parallel processes of interactive learning from testing possible combinations. Six facilitating mechanisms to be explored further were identified in this exploratory study.

Keywords. Transdisciplinary innovation, Concurrent engineering, Bricolage, Community engagement, social capital

Introduction

Innovation is a social process involving a number of actors - it is people who come up with ideas and promote them. A colleague once likened the evolution of an innovation to making a movie. Whilst the process has a beginning and an end, the script may be adapted along the way and some scenes may have to be filmed several times. Some of them may never appear in the final cut. The plot will have twists and turns, and whilst there will be a few central actors, as shown in the credits at the end, there will be support from many others having particular specialisations. A particular film will be framed and shaped by considering the intended audience.

Consistent with this analogy, a study of research methods in theatre and performance by Kershaw et al [1] describe transdisciplinary innovation as 'practice as research' associated with creative enquiry. In reviewing approaches to transdisciplinary research in the health sector, Kessel and Rosenfield [2:S225] observed that "far-sighted [World Health Organisation] medical doctors and international health workers began to realize that the effective delivery of health care, especially in cross-cultural settings, involved sociocultural as well as purely medical factors." What we take from this is that the concept of transdisciplinary innovation involves the melding of social, technological and organisational perspectives oriented towards the needs/wants of the

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intended beneficiaries. Some see this as requiring a new thought style for the innovators involved (e.g. Darbellay [3]), and as being a powerful tool in developing economies (e.g. Schensul, Nastasi, & Verma [4]). Our review of the literature identified three contexts in which the term transdisciplinary innovation was used, as outlined in Table 1.

Table 1. Some issues associated with transdisciplinary innovation identified in the academic literature.

References	Transdisciplinary Innovation Context
Hadorn et al [5]; Nowotny, Scott, & Gibbons [6], Stokols et al [7]; Gray [8]; Klein [9]	Issues related to the utility and practice of transdisciplinary research
Boujut and Laureillard [10]; Sobolewski [11]	Issues related to implementing the design of an innovative new product or service
Emmons et al [12]; Schensul et al [13]; Keating et al [14]	Issues emerging in confronting a real-world complex problem where there were multiple stakeholder interactions

We suggest that the process of transdisciplinary innovation involves all three aspects, as illustrated in figure 1.

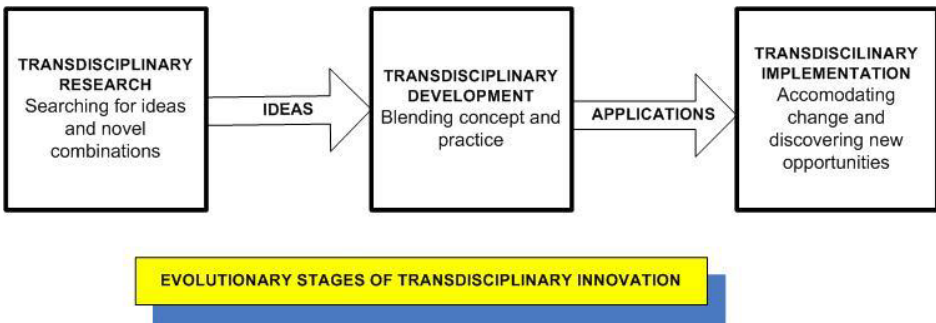


Figure 1. Transdisciplinary innovation model.

In this paper we reflect on what transdisciplinary innovation might look like in practice in three joint Australian - Indian academic cases. We explore the research question ‘what mechanisms might facilitate transdisciplinary innovation’, drawing on inferences from our case study analysis.

1. Research Approach

A longitudinal multiple case study approach based on Yin's [15] methodology was used to establish rich data sets covering all stages in the evolution of an innovation and preserving context at this exploratory stage of the research. The purposive sample of cases sought to cover a variety of situations. The literature had suggested that transdisciplinary innovation may be enacted differently depending on technological and community context. We draw on three cases involving collaboration between an Australian and Indian academic institution which considered topics of national importance. One had a business outcome orientation, and two had a social outcome orientation.

Cross case analysis was conducted using Eisenhardt's [16:540] tactic of selecting a particular dimension, and considering similarities and differences between cases in the

context of that dimension. The dimension chosen was the evolutionary stage of a transdisciplinary innovation illustrated in figure 1. Each of the ten references cited in table 1 was examined to identify a central theme, and observations from the cases were compared in relation to each theme. Some themes related to matters of agency and some to matters of structure. By way of example, Gray [8] considered leadership challenges and collaboration competencies. In this context, an innovation champion role and the value of social networking were observed in the cases. The process provided thirty observations about some aspect of transdisciplinary innovation. The results were tabulated to facilitate comparison, but space limitations precluded their presentation here. A brief outline of each case data set follows.

1.1. Case 1: Food Sector SMEs: Feeding Growing Populations

This data set came from a collaboration between an Australian and an Indian university exploring the dynamics of innovation in food sector SMEs. The original research question was: what are the sources of innovation for SMEs in the food products manufacturing industry? Both governments regarded growth in this sector as important. Secondary data from websites, company documents and news media plus direct senior executive interview data was collected from seven SMEs in Australia and five SMEs in India that had been in business more than 10 years. Representatives of a regional cluster were interviewed in each country. This data set was analysed to explore the nature of the innovation process at work and business sustainability mechanisms. Food security and quality were community concerns. We specifically asked the enterprises studied in Case 1 about their use of transdisciplinary/open innovation practices in considering such concerns whilst maintaining economic sustainability. They did not recognise or use the formal processes described in the literature. They collected external ideas via personal networking with regional community groups, communities of practice and trade events. They not only sought ideas for a product innovation, but ideas about process innovation and what customers valued. Innovative products were developed from novel combinations of readily available ingredients and established practices, homing in on 'recipes' via a process of trial and error. Business models changed in response to market pressures / opportunities.

1.2. Case 2: Deploying Vision Science to Confront a National Health Problem

This data set provided information on the cumulative effect of multiple prior collaborations and project management approaches aimed at deploying radical eye treatment innovations where proof of concept had been demonstrated in the laboratory. The Indian partners were experienced eye surgeons, an enterprise that could manage extensive patient field trials in India, and an industrial firm that might make novel implants. Data was collected from private meetings with the Australian researchers over a six-month period and participation in a one-week partner workshop involving medical and technological experts where suitable project management and initial production arrangements for the complex technologies involved were explored. A multi-national researcher-practitioner in-kind collaboration was established to confirm the product practicality of two new research concepts through pilot scale production and field trials. It was intended a start-up company funded by venture capital be established to manage commercialization, but problems emerged. The Global Financial Crisis impacted the availability of venture capital.

1.3. Case 3: Kerala Disaster Management: Pre-emptive Action to Avoid Disasters

This data set related to a collaborative project involving Amrita University in India teaming with some international universities, also involving a consulting firm (the Sustain Group), and the local community that had been financed by Amrita. A natural disaster had struck a particular village and, as a result, there was a significant land shift at the top of a pass that hung over many of the houses. The local community was primarily engaged in the tea industry and many villagers were living on or below the poverty line, so they could not easily relocate. Amrita University worked with the local community and the Sustain group to install high technology measurement equipment that enabled data to flow directly back to a team to assess land movement and make predictions, enabling the local community to move to safer ground in the event of another impending disaster.

An early warning system was deployed in a landslide prone area where two major landslides had occurred. Eighty geological sensors in twenty deep earth probes collected and transmitted data to the Amritapuri Data Analysis Center using a wireless sensor network of twenty-five nodes and satellite connections, as establishing a hard-wired system in the area was impossible. Large-scale data collection and power management algorithms facilitated management of the system. Data has been continuously transmitted from the deployment site in real time to the university since 2008.

2. Observations from the Cases

Multiple disciplines were involved in all cases:

- In case 1 this included agricultural science, food science, production engineering, logistics and marketing, much of which was combined in a business model - a strategy for the delivery and extraction of value.
- In case 2 this included polymer science, manufacturing engineering and medical science
- In case 3 this included sensor science, systems engineering, geo-physical science, and data processing/interpretation

There was a common focus on interaction mechanisms. Whilst cases 2 and 3 had a traditional scientific heritage, the implementation of that research also involved significant field learning: learning *for* innovation and learning *from* innovation in conjunction with users/beneficiaries.

Six recurring themes emerged from the cross-case analysis outlined earlier, and these were viewed as mechanisms facilitating transdisciplinary innovation. Table 2 outlines each mechanism with reference to associated literature.

Table 2. Mechanisms facilitation transdisciplinary innovation.

Facilitating Mechanism	Observations from the cases
The influence of Social Networking	Purposeful social networking was a common practice of the SME firms in case 1. Different networks were accessed for different purposes - some to passively receive information, some to actively broadcast information, and some for intimate exchanges of information. In case 2, the latter was facilitated by project workshops, whilst broadcast information was held in an on-line knowledge management system. Third parties were sometimes involved in facilitating network connections. In case 3, the Indian university had established professional connections within the international science community and regional community engagement connections.
Network Thinking and Interface Protocols	Whilst the literature refers to arrangements for organising transdisciplinary teams, we suggest it is more appropriate to think of transdisciplinary networks of autonomous agents and interfaces to enable them to interact. This recognizes that individual actors are also connected to external actors who may indirectly contribute to a transdisciplinary project. For example a biomedical engineering practitioner may have links to biology, medical and engineering scientists. Who is connected to who, what is connected to what, and what protocols facilitate communication? Industry associations provided strong connections in case 1. A strong orientation towards managing interfaces was observed in cases 2 and 3 both in relation to the product/system under development and in relation to management arrangements. This way of thinking recognizes that particular actors may be linked in different ways at different times. For example, in case 2 the field trials actors had a limited involvement until regulatory approval was obtained to begin trials.
Boundary Objects and User Stories	In case 1 interface artifacts were prototype products and processes constructed to get user feedback in an iterative style of development. In case 2 early prototypes were used to understand what was possible before moving on to what was practical. Face-to-face workshops were used to share knowledge and provide an opportunity for critical questioning. In case 3 funding bodies required regular reviews and reports, which caused the participants to document their viewpoints. Boujut and Laureilland [10] raised the matter of interface artifacts, interface roles and interface knowledge that facilitated cross-discipline engagement in a design setting, and related observations. Star and Griesemer [17] refer to boundary objects where the viewpoints of both expert and community contributors from different backgrounds were pooled in considering an archeological artifact - what was it, what could it be used for, and how was it made? A story related to the artifact emerges, and can be discussed and elaborated on. Having cohesive user stories about their needs is an important starting point in agile software design [18], and can represent a desired future if a complex problem is well framed.
The Utilisation of Backcasting	Backcasting is seen as a practice of imagining future options applied to long-term complex issues (as seen for example in our cases 1 and 2), then working out what has to be in place to realize a particular future (e.g Dreborg, [19]). For example, how can we achieve the goal of providing safety for people living in a landslide-prone area (Case 3). The next question is how to put the supporting infrastructure in place and how to utilize it. Can we combine current knowledge in some way, or do we need to develop new knowledge? What kinds of knowledge are needed (e.g. system, target or transformational knowledge - Hadorn et al [5]) This led to a variety of R&D activities, some within a particular discipline and some transdisciplinary.

Facilitating Mechanism	Observations from the cases
Unifying Transdisciplinary Knowledge Elements	Madni [20] saw transdisciplinary innovation as bringing together elements of knowledge from disparate sources. Hargadon [21] has suggested: "it is the recombinant (rather than inventive) nature of revolutionary innovations that contribute to their dramatic effects". Weik [22] introduced the idea of improvisation as a mindset associated with creativity and innovation, spontaneously putting together combinations of ideas, some of which succeed and some of which fail. Baker and Nelson [23] refer to "creating something from nothing: resource construction through entrepreneurial bricolage" - drawing together whatever resources may be readily available to pursue a current need or opportunity. There are some common themes here - being aware of or assembling a parcel of knowledge elements, experimenting with combinations of them, appreciating the utility of a particular combination, and building an expanded resource platform to support further innovation. This is illustrated in the evolution of mobile phones from telephones to messaging devices, to the inclusion of image capture, to the incorporation of computing and GPS capabilities. Albeit in a different way, this kind of journey has also been taken by our case 1 firms. In case 2, the participants were building on what was learned in a successful prior project [24]. In case 3 Australian, European and Indian participants with particular expertise were drawn together
The Interplay of Two Modes of Innovation	It has been suggested that: "The old paradigm of scientific discovery (Mode 1) characterised by the hegemony of disciplinary science, with its strong sense of an internal hierarchy between the disciplines and driven by the autonomy of scientists and their host institutions, the universities, is being superseded - although not replaced - by a new paradigm (Mode 2). This mode is socially distributed, application-oriented, transdisciplinary and subject to multiple accountabilities" [6]. We will call these discovery mode and interaction mode respectively. Discovery mode was rarely observed in our case 1 firms, but was an important element of innovation in cases 2 and 3. In case 1, interactive mode multiple influence factors related to shareholders and Corporate Social Responsibility undertakings, recognizing that most of the firms involved were embedded in their local communities.

3. Concluding Remarks

Exploring the academic literature using the keywords transdisciplinary and innovation identified studies framed in three different contexts: in research practice, in product/process development and complex community problem-solving. We contend that all three are combined in the evolution of a transdisciplinary innovation, as illustrated in the model shown in figure 1.

Drawing on longitudinal case studies that embodied these three contexts in different ways, we identified six facilitating mechanisms (see Table 2). Social networking between both professional and regional communities was seen as important, along with ways of facilitating interaction between these communities such as the adoption of common interfaces, including language and jargon, and the use of boundary objects. Having a broad long-term common goal supported each discipline member of multiple social networks to identify how they could contribute to achievement of this goal. A focus on multi-stakeholder outcomes had the effect of unifying transdisciplinary knowledge elements.

It had been suggested in the literature that transdisciplinary innovation required a different, interactive research approach, described as Mode 2 by Nowotney et al [6]. In

our cases, this supplemented rather than replaced discipline-based discovery research, which provided new ideas that could be combined with other ideas.

It has been suggested that transdisciplinary innovation emerges from the combination of diverse knowledge elements. In this regard, we like the observation of Kanter (25:73) that "Innovation is always a surprise. By definition, it is something no-one has thought of before. Its very existence shows that reality is not fixed in predictable patterns. Instead creative new possibilities can emerge in any field, in any industry. Innovators see new patterns in the familiar, apparently immutable, situations. It is though they see the world through a kaleidoscope, which creates endless variation from the same set of fragments".

Using this analogy, both turning the kaleidoscope and changing or adding elements to it can yield quite different patterns. In our cases, social networking facilitated the identification of new knowledge elements and new viewpoints.

We did not come up with a design for a transdisciplinary innovation kaleidoscope, but we identified six questions that might be asked in contemplating transdisciplinary innovation initiatives:

- What social networks (professional and community) need to be engaged to pursue a particular innovation - what exists and what needs to be created?
- Who are the network nodal actors and what interface protocols have to be in place for their effective interaction?
- What kinds of 'boundary objects' and 'user stories' are needed to facilitate common understandings?
- How can backcasting techniques be used to realize a unifying, but broadly stated goal?
- How can we identify, share and unify transdisciplinary knowledge elements?
- What might be the appropriate interplay between two models of research, characterised as discovery and interaction?

These questions might be asked in relation to each stage in the evolution of an innovation illustrated in figure 1.

One can only draw limited conclusions from three case studies, and whilst there were interactions between multiple disciplines in all cases, the extent to which they deliver transdisciplinary innovations requires further investigation. Insights from the cases suggest this occurs at the micro-level, but further study is beyond the scope of this paper. The model shown in figure 1 and the six interaction mechanisms identified provide a foundation for future research.

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